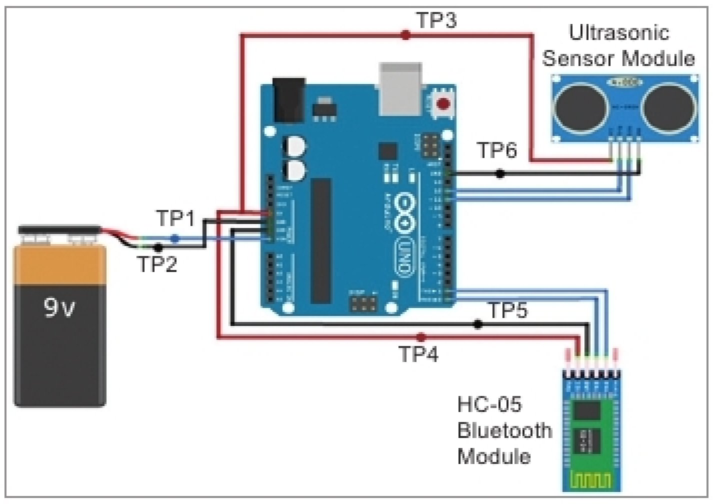

Fig. 5: Connection diagram of the project (transmitter)

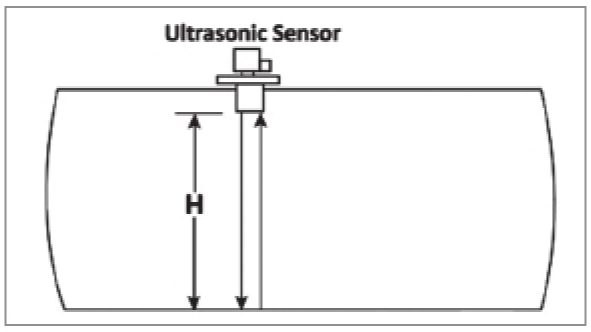

Fig. 6: Empty tank

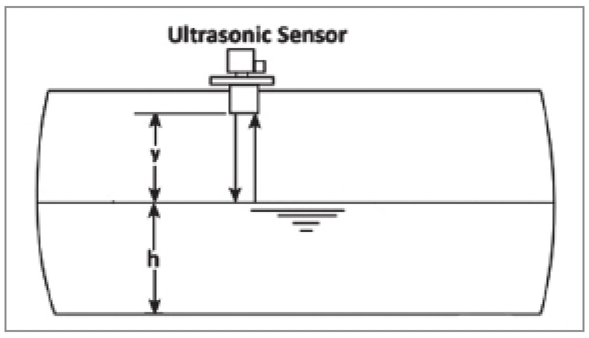

Fig. 7: Tank filled with water to a certain height (h)

Fig. 8: Water level (9cm) displayed on smartphone

tion with accuracy up to 3mm. The module includes an ultrasonic transmitter, receiver, and control circuit.

It has four pins—echo, trig, Vcc, and GND. The basic working principle of the module is as explained here.

Trig(trigger) pin is an input pin. It is made for a high pulse of at least 10µs duration, on receipt of which it automatically sends eight sonic bursts of 40kHz (ultrasonic signals). This signal, after reflecting from the object/water, is received by the echo pin.

The echo pin is an output pin. It goes high for a period of time, which is equal to the time taken for the ultrasonic wave to return to the sensor. The echo pin provides the travelling time of the sound wave in microseconds.

The connection diagram of the ultrasonic module with Arduino is shown in Fig. 5.

The ultrasonic sensor measures the distance between the sensor and object/liquid by sending an ultrasonic wave and calculating the time required for the echo to return. If the time between the trigger and echo is known, the distance of the object can be calculated by simply multiplying half of the time with the speed of sound in air. This principle is used here to detect the level of water in a tank. The working of the sensor is illustrated in Figs 6 and 7.

As shown in Fig. 6, when the tank is empty, the ultrasonic wave hits the bottom of the tank and gets reflected. As a result, it detects the depth of the empty tank at 'H' height.

If the tank is filled with water up to 'h' height, the ultrasonic wave is reflected from the water surface, and it detects the depth of empty space 'y' above the water, as shown in Fig. 7.

When 'H' and 'y' are known, the water level 'h' can be calculated by subtracting 'y' from 'H,' that is, h = (H-y). The value of H is calculated only once during the initial calibration when the tank is empty. For our setup, the value of H was 17cm. You can change the value of H in the code to 30cm, 50cm, or any other value depending on the height of the tank.

The value of the water level is sent to the HC-05 Bluetooth module using serial communication. As shown in Fig. 5, RXD and TXD pins of the Bluetooth module are connected to TX and RX pins of the Arduino Uno board, respectively. The HC-05 Bluetooth module sends this data wirelessly to the smartphone via Bluetooth. The data received on the smartphone is displayed on the app. To connect the Bluetooth module with the smartphone, these must be paired.

EFY Note
The source code of this project is available for free download at *source.efymag.com*

Table 1: Expected Voltages at Different Test Points (Refer Fig. 5)

Test point	Expected voltage
TP1	+9V
TP2	GND (0V)
TP3	+5V
TP4	+5V
TP5	GND (0V)
TP6	GND (0V)

Construction and testing

Download the 'smart_water_level.ino' source code to the Arduino Uno board. The 'Level Monitoring' application is installed on the Android smartphone. Next, turn on the Bluetooth on your smartphone.

When you power on the circuit (transmitter side), the HC-05 Bluetooth module gets paired with Bluetooth on your Android smartphone. The default passkey is '1234' or '0000'.

After successful pairing, open the 'Level Monitoring' app from the smartphone and press the Bluetooth logo. You will get a list of Bluetooth devices detected by your smartphone. You need to select an HC-05 device from the list.

After a successful connection, you will get a 'Connected' message on the main screen of the application. The Android application will now start to display the water level data received from the transmitter side.

Assemble the circuit, as shown in Fig. 5, and place the ultrasonic sensor at top of the vessel/container whose liquid level you want to measure. For troubleshooting, you can check voltages at different test points, as shown in Table 1.

A screenshot of the smartphone displaying the water level = 9cm during testing is shown in Fig. 8. EFY


Saikat Patra is passionate about electronics and MCU based embedded system applications


Shibendu Mahata is M.Tech (Gold Medallist) in Instrumentation and Electronics Engineering from Jadavpur University. He loves designing MCU based real-time embedded signal processing and process control systems